

Prompt Engineering - Student Level Mini Projects

Quick project titles and implementation flow architecture aligned with the syllabus.

1. AI Resume Builder

Architecture Flow:

User Details → Prompt Template → LLM → Resume → PDF

2. Smart Email Generator

Architecture Flow:

Topic → Role Prompt → LLM → Professional Email

3. AI Study Notes Generator

Architecture Flow:

Topic → Prompt → LLM → Summary + MCQs + Notes

4. AI Quiz Generator

Architecture Flow:

Topic → Prompt → LLM → JSON Questions → Quiz

5. Prompt Comparison System

Architecture Flow:

Question → Zero-shot/Few-shot → Compare Outputs

6. Student PDF Chatbot (Mini RAG)

Architecture Flow:

PDF → Chunking → Embeddings → Vector DB → Retriever → LLM

7. College FAQ Chatbot

Architecture Flow:

College Docs → Vector DB → Query → Retriever → LLM

8. AI Image Caption Generator

Architecture Flow:

Image → Vision Model → Caption

9. Text-to-Image Prompt Designer

Architecture Flow:

Idea → Prompt Optimizer → Image Model

10. AI Travel Planner

Architecture Flow:

Inputs → Prompt → LLM → Itinerary

11. AI Interview Assistant

Architecture Flow:

Role → Prompt → LLM → Questions + Feedback

12. AI Calculator Agent

Architecture Flow:

User Query → Agent → Calculator Tool → Answer

13. Prompt Evaluation Dashboard

Architecture Flow:

Prompts → Outputs → LLM Judge → Scores

14. Prompt Injection Detector

Architecture Flow:

User Prompt → Safety Check → Allow/Block

15. AI News Summarizer

Architecture Flow:

Article → Prompt → LLM → Summary

16. AI Document Analyzer

Architecture Flow:

PDF → Extract Text → LLM → Insights

17. AI Language Translator

Architecture Flow:

Text → Prompt → LLM → Translation

18. AI Meeting Minutes Generator

Architecture Flow:

Meeting Notes → LLM → Minutes

19. AI Code Reviewer

Architecture Flow:

Code → Reviewer Prompt → LLM → Suggestions

20. AI Learning Assistant (Capstone)

Architecture Flow:

Streamlit UI → Prompt → RAG/Agent → LLM → Response